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Saveetha Institute of Medical And Technical Sciences

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# Fresh AI-Assisted Phishing Email Detection and Secure Communication Framework

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**CSA5111 - CRYPTOGRAPHY AND NETWORK SECURITY**


Guided By:

Dr. Sivasankar chandrasekaran

Presented By :

CHALLA RAVITEJA (192325065)

TALAPALA PARTHA SARATHI (192325042)

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# Introduction

- Email is widely used for personal, academic and professional communication.
- Phishing emails trick users into revealing passwords, financial information or other sensitive data.
- Traditional email filters may fail to identify newly created or sophisticated phishing messages.
- Artificial Intelligence (AI) and Machine Learning (ML) can analyze email content and patterns to classify potentially malicious messages.
- Pretty Good Privacy (PGP) can support email encryption and digital signatures for confidentiality, integrity and sender authentication.

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# Problem Statement

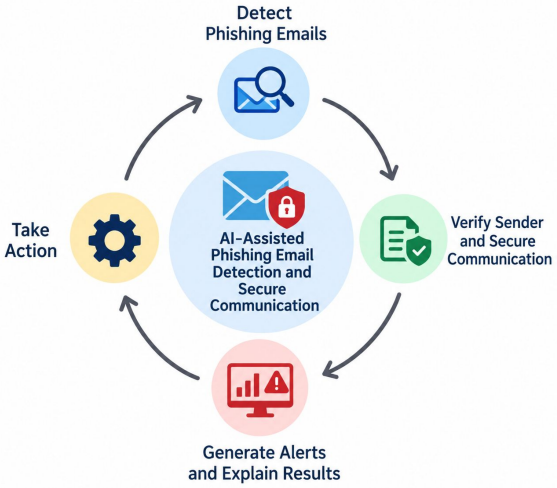
Existing email communication systems may struggle to detect sophisticated phishing attacks while providing secure communication and understandable explanations for suspicious messages. Therefore, a framework is needed to combine AI-assisted phishing detection, PGP-based email security and explainable threat alerts.

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# Objectives

The main objectives of this project are:

- To prepare email datasets and extract meaningful features for phishing detection.
- To develop an ML-based model to classify emails as phishing or legitimate.
- To demonstrate secure email communication using PGP encryption and digital signatures.
- To provide understandable explanations for phishing predictions.
- To display classification results, security status and alerts through a dashboard.



# Literature Review

| S.No | Year | Reference | Related Work Summary | Tools / Tech | Data / Components | Feasibility Study |
|---|---|---|---|---|---|---|
| 1 | 2024 | Interpretable Phishing Detection Platform | Detects phishing emails with explainable AI. | ML, XAI | Email datasets | Improves prediction transparency. |
| 2 | 2025 | Desolda et al., APOLLO | Detects phishing and explains threats. | GPT-4o, NLP | Email content | Provides understandable warnings. |
| 3 | 2024 | RFC 9580, OpenPGP | Defines email encryption and digital signatures. | OpenPGP | Keys, messages | Supports secure communication. |
| 4 | 2025 | Tooher and Lallie | Uses deep learning to detect phishing. | Deep Learning, NLP | Labeled emails | Helps identify suspicious messages. |

# Existing System and Its Limitations

**Existing System**

- Spam filters use predefined rules and sender reputation.
- Some systems use Machine Learning to classify phishing emails.
- PGP tools provide encryption and digital signatures.
- Email security tools may display warnings about suspicious messages.

**Limitations**

- New phishing emails may bypass existing filters.
- Some ML models do not clearly explain their predictions.
- Encryption alone does not establish whether an email is malicious.
- Sender addresses may be spoofed.
- Detection and secure communication may be handled by separate tools.

# Proposed System and Advantages

**Proposed System**

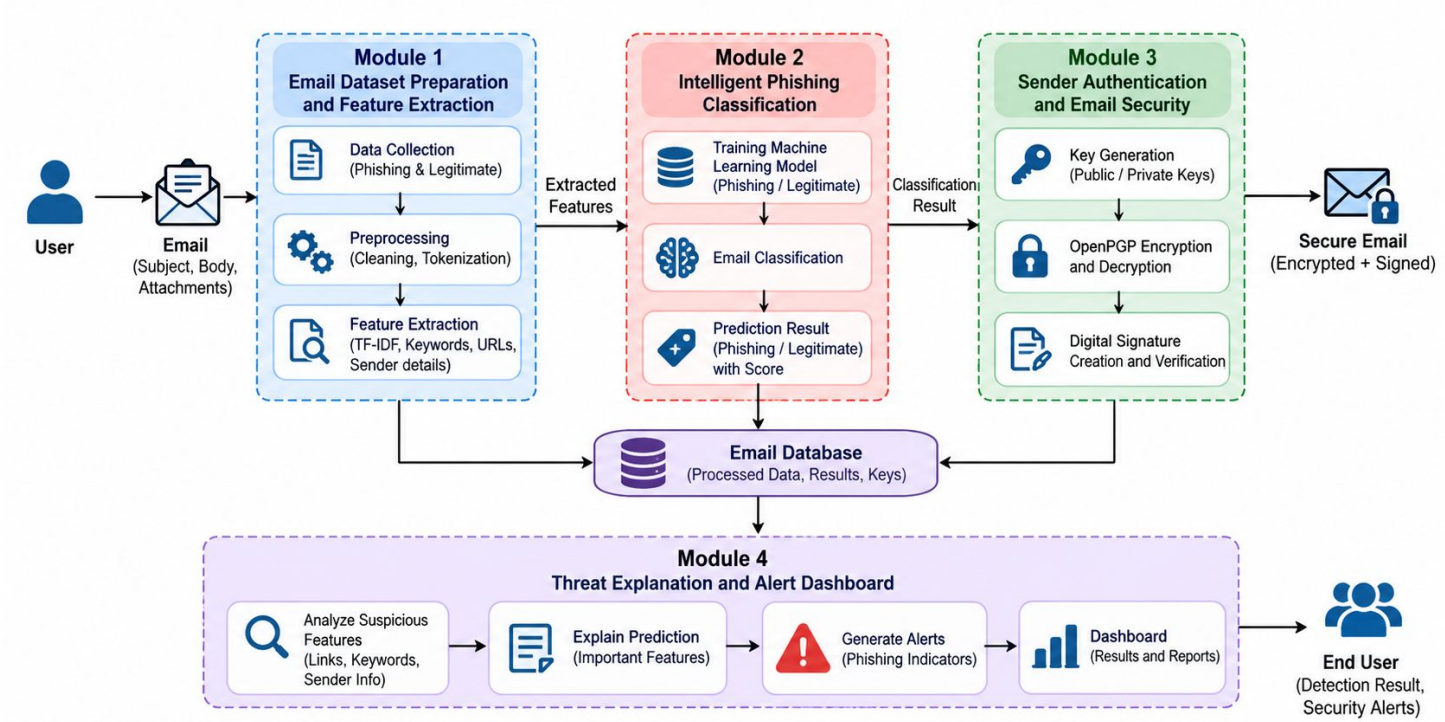
The proposed framework integrates AI-based phishing detection, PGP-based secure communication, and explainable alerts in one web application.

**Advantages**

- Classifies emails as phishing or legitimate.
- Uses email content and selected metadata for classification.
- Supports encrypted email communication and digital signatures.
- Explains suspicious indicators to the user.
- Displays classification results and security alerts through a dashboard.
- Evaluates performance using standard classification metrics.

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# System Architecture



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# Modules

**Module 1:** Email Dataset Preparation and Feature Extraction

**Module 2:** Intelligent Phishing Classification

**Module 3:** Sender Authentication and Email Security

**Module 4:** Threat Explanation and Alert Dashboard

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# Module 1: Email Dataset Preparation and Feature Extraction

**Purpose:** Prepare email data and extract relevant features for phishing detection.

**Functions:**

- Collect phishing and legitimate email datasets.
- Clean and preprocess email subject and body text.
- Extract keywords, URLs, sender details, and suspicious patterns.
- Convert email text into numerical features using TF-IDF.

**Input:** Raw email dataset containing phishing and legitimate emails.

**Output:** Cleaned dataset with extracted features ready for classification.

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# Module 2:
# Intelligent Phishing Classification

**Purpose:** Identify whether an email is phishing or legitimate using machine learning.

**Functions:**

- Train a machine learning model using labeled email data.
- Apply the trained model to the extracted email features.
- Classify emails as phishing or legitimate.
- Generate a prediction score for each analyzed email.

**Input:** Processed email features from Module 1.

**Output:** Email classification result with a prediction score.

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# Module 3:
# Sender Authentication and Email Security

**Purpose:** Protect email communication and verify digital signatures.

**Functions:**

- Generate or obtain public and private key pairs.
- Encrypt messages using OpenPGP when secure communication is required.
- Decrypt encrypted messages using the authorized private key.
- Verify digital signatures to check message integrity and key authenticity.

**Input:** Email message, encryption keys, and digital signature.

**Output:** Secure message with encryption and signature verification status.

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# Module 4:
# Threat Explanation and Alert Dashboard

**Purpose:** Display phishing detection results, explain suspicious features, and generate security alerts.

**Functions:**

- Display the classification result and prediction score.
- Highlight suspicious keywords, links, and sender information.
- Explain the important features influencing the model's prediction.
- Generate alerts and display analysis results through a dashboard.

**Input:** Classification results, extracted features, and email security status.

**Output:** Threat explanation, security alerts, and visualized email analysis results.

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# Module-wise Inputs and Outputs

| Module | Input | Processing | Output |
|---|---|---|---|
| Module 1 | Raw email dataset, subject, body, sender details | Data preprocessing and TF-IDF feature extraction | Cleaned email data and extracted features |
| Module 2 | Extracted email features | Machine learning-based phishing classification | Phishing/legitimate prediction and score |
| Module 3 | Email message, public/private keys, digital signature | OpenPGP encryption, decryption and signature verification | Secured message and signature verification status |
| Module 4 | Classification results, suspicious features, security status | Threat explanation, alert generation and dashboard visualization | Threat alerts, explanations and detection report |

**Common data flow:**

The output of Module 1 becomes the input to Module 2. Module 3 performs encryption, decryption and digital-signature verification when secure communication is required. Module 4 combines the classification results, suspicious indicators and security status to display understandable explanations, alerts and a detection report.

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# Technologies, Tools, Hardware and Software Requirements

| Component | Technology |
|---|---|
| Programming language | Python |
| User interface | Streamlit |
| Data processing | Pandas, NumPy |
| Machine Learning | Scikit-learn |
| Text feature extraction | TF-IDF |
| Explainable AI | Model coefficients or LIME |
| Email security | GnuPG/OpenPGP tools |
| Data visualization | Matplotlib, Plotly |
| Development environment | VS Code |
| Operating system | Windows 10/11 |
| Optional storage | SQLite |

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# Output

- An AI-based **phishing email detection system** classifies emails as phishing or legitimate.
- **Email preprocessing and feature extraction** identify suspicious keywords, URLs, and sender information.
- **Secure email communication** supports OpenPGP encryption, decryption, and digital signature verification.
- **Threat explanations and security alerts** help users understand suspicious email indicators through a dashboard.
- **Classification results and evaluation metrics** provide insights into the system's detection performance.

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# Future Scope

- Integrate **Deep Learning models** to improve phishing email detection accuracy.
- Extend detection to **QR-code phishing, malicious attachments, and advanced phishing URLs**.
- Implement **real-time email monitoring** with automated threat alerts.
- Enhance **Explainable AI (XAI)** to provide clearer reasons for phishing predictions.
- Integrate **cloud-based security services** for scalable email analysis and threat intelligence.
- Develop **browser extensions and mobile applications** for wider cybersecurity protection.

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# Conclusion

The Fresh AI-Assisted Phishing Email Detection and Secure Communication Framework provides an integrated approach to identifying phishing emails and improving email security. It combines machine learning techniques to classify emails as phishing or legitimate with OpenPGP encryption and digital signature verification for secure communication. The system also provides explainable predictions, highlights suspicious email indicators, and generates security alerts through a centralized dashboard. Overall, the proposed framework aims to improve email protection by combining intelligent phishing detection, secure communication, and understandable threat analysis.

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# Thank you!